

Minutes of the Meeting of the 21st Central Facilities Committee (CFC) held on 20.10.2023

The 21st CFC meeting was held in the Boardroom of IIT Delhi at 11:00 hours on the 20th of October 2023. It was attended by the following members: Prof. Ambuj Sagar (Chairperson, DDSP), Prof. Srikrishnan (DDO), Prof. Vivek Buwa (Dean Planning), Prof. Naresh Bhatnagar (Dean RND), Prof. Bhanu Nandan, Prof. Neetu Singh, Prof. Nityanand Goswami, Prof. Pankaj Srivastava (Ex Head CRF), Prof. Sameer Sapra (Head CRF) and Prof. Manidipa Banerjee (Convenor, Associate Head CRF).

The following agenda items were discussed.

1. Confirmation of the minutes of the 20th CFC meeting: The minutes of the 20th CFC meeting were confirmed as circulated.

2. Activities of CRF in the last 1 year: The Head CRF briefly appraised the committee regarding the activities of the CRF in the last year, including commissioning of new facilities and those in the pipeline, workshops and student training, the status of the CRF-SATHI building at the Sonipat campus and other initiatives.

3. Formulation of Policy for CRF Employee Benefits: The committee deliberated on benefits, such as leave, bonus, HRA, DA, TA, medical booklet, accommodation, pension plan etc., for staff who have been employed in CRF for more than 5 years. It was suggested that since any decision will have major financial implications, as well as implications for staff in other departments, an institute-wide policy needs to be established, and a corpus must be created for such activities. Therefore, no decision regarding CRF staff can be taken at present and will have to await discussions with the Director and BoG, and the framing of the aforementioned policies.

4. Honorarium policy for CRF employees: The CFC decided that honorarium may be paid to CRF staff for specialized services such as operating facilities and assisting users beyond regular working hours or for training programs/workshops. The amount may be recommended by the faculty coordinator of the facility/Head CRF/Associate Head CRF and approved by Head CRF.

5. Sustainability of CRF employees: The CFC noted that the experience and expertise of CRF staff are crucial in order to maintain a cutting edge technical/scientific platform and generate high quality research data, and to retain such highly trained staff and prevent attrition is essential. The current recruitment rules will not allow shortlisting of staff previously hired in contracts for PTO/TO positions as these positions require 2 years of work experience in a regular position. The CFC requested a roadmap for permanency of such employees from the Deputy Director Operations. The DDO emphasized that while hiring of technical staff in entry level positions for CRF operations may be possible, hiring of specialized staff in PTO/TO positions will require reframing of the RR and is not immediately feasible. Hence, this matter also needs further discussions for implementation.

6. Five-year budget projection for CRF operations: The Head, CRF presented a budget projection for the next five years to the CFC (*Annexure I*). The CFC strongly recommended that in addition to current funding modes, additional modes should be explored for fund generation, including active solicitation of industry participation (also discussed in agenda item 9). CFC also suggested exploring permanent position for long-serving CRF manpower (also discussed in agenda item 5).

7. Approval of remaining equipment to be procured through HEFA funds: The CFC approved the acquisition of the following three equipment from HEFA funds:

- a) Helium liquefier (Rs 2.8 Cr)
- b) Liquid N₂ plant (Rs 2.5 Cr)
- c) E motor dyno (Rs 2 Cr)

A final list of the equipment purchased/to be purchased through HEFA funds is attached as *Annexure II*.

8. Approval of proposals to be funded through CRF: The Head, CRF informed the CFC about the process utilized to prioritize equipment/facilities to be purchased. Briefly, a campus-wide call was launched for proposals for Central Facilities, and 30 were received. These proposals were scrutinized through presentations to a committee consisting of CFC members, as well as faculty members nominated by the Heads of Departments/Schools/Centers. The presentations were also made accessible to all faculty of IITD. Based on the proposal content and presentations, ten facilities have been shortlisted by the committee (*Annexure III*). The CFC suggested circulating these to all faculty, to receive input on the following:

- a) Potential users of the facility
- b) Time required by each user on the facility
- c) User publications/research grants supporting the usage of a particular facility
- d) Whether the user is willing to commit funds/write grants to support the facility

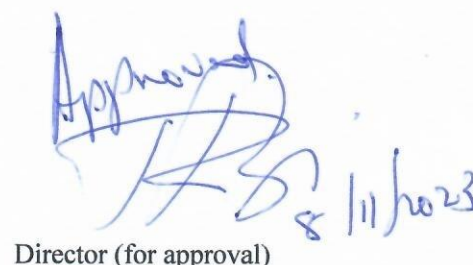
The CFC emphasized that no more than 2 such facilities may be supported by each faculty member. A priority list for purchase of these equipment will be made by the CFC once the potential usage data is available. The CFC also suggested that GTE exemption requests and other purchase formalities should be initiated for these facilities.

9. Sustainable model for CRF operations: The CFC suggested that in order to make the CRF facilities sustainable in the long run, it is essential to place equipment/facilities into clusters as per scientific usage and under the coordinator-ship of a group of core faculty. This core group will be encouraged to augment existing funds by generating additional funds through central govt agencies, industry partners or academic institutes. The CFC strongly emphasized that additional funds generation by interested users is essential for long term viability of the CRF platform.

The meeting ended with a vote of thanks to the Chair.


Head CRF 02/10/23


Chairman, CFC (DD S&P) 6/11/2023


Director (for approval) 8/11/2023

All CFC members

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HOD, CRF

Annexure I**Next 05-year CRF projection**

Financial Year	NPN Budget	PLN Budget	Salary/Manpower
FY 23-24	2.0 crores	3.5 crores	2.5 crores
FY 24-25	3.0 crores	3.5 crores	3.0 crores
FY 25-26	4.0 crores	4 crores	3.5 crores
FY 26-27	5.0 crores	4 crores	4.0 crores
FY 27-28	6.0 crores	5 crores	4.5 crores
FY 28-29	7.0 crores	5 crores	5.0 crores
	* AMCs/CMS for 19 more equipment		*Increased by 50 lakhs every year based on the annual increment of existing staff. *And addition of 15 more staff in next 5 years

Annexure-II**List of HEFA Equipment**

S.No.	Name of Instrument	PR Amount	Payment status
1.	Ion Milling System	2,49,96,400	Payment Done
2.	Femto second laser	4,32,29,760	Payment Done
3.	STM	6,46,386	Payment Done
4.	PPMS	5,81,57,800	Payment Done
5.	TOF-SIMS	7,56,41,080	Payment Done
6.	Helium cooled Cryoprobe for 500MHz NMR	2,17,26,694	20% left
7.	AMC order Helium cooled cryoprobe	66,33,030	20% left
8.	EPR	4,44,99,208	Payment Done
9.	BET Surface Area Analyzer	95,96,998	Payment Done
10.	High Resolution DHM	39,99,975	Payment Done
11.	Nanoparticle Tracking System	64,76,080	Payment Done
12.	CD Spectrometer	2,26,56,048	Payment done
13.	AECL	14,16,53,703	Payment Done
18.	400 MHz NMR	3,51,78,000	Payment Done
19.	Test Bench for EV motor simulator	2,95,12,082	Payment Done
20.	E Motor Dyno	2,00,00,000	To be tendered
21.	JEM ARM HRTEM accessories	2,58,72,000	20% left
25.	Helium Liquefier	2,80,00,000	To be tendered
26.	Cryostat	1,80,32,000	20% left
27.	Liquid Nitrogen plant	2,50,00,000	To be tendered
	TOTAL	64,15,07,244	

Annexure-III**List of the Proposals in Agenda 8**

S.No.	Facility Name	Cost of the Facility
1.	High throughput protein Characterization & binding facility	4.0
2.	Flow measurement & characterization facility	3.5
3.	Peptide synthesis & purification facility	1.82
4.	MPMS3 SQUID Magnetometer facility	8.0
5.	Stable isotope ratio mass spectrometer (IRMS) facility	4.5
6.	Zebrafish facility	2.0
7.	Accelerated rate calorimeter/thermal and abuse characterization of batteries facility	2.0
8.	Versatile ultra-low & high load nano indentation facility	5.16
9.	Automated Electrophysiology facility	1.44
10	FESEM with FIB	8.0
	TOTAL	40.42